

Synthetic Fixed-Confidence Experiments for Binary Pairwise Pareto Set Identification

Experiment report generated from `Exp_Synthetic`

July 8, 2026

Abstract

This report summarizes the completed synthetic experiments for fixed-confidence binary pairwise Pareto set identification. The experiments compare VB-EGE-practical against three fixed-confidence benchmarks: UniformFocalBorda-FC, UniformPairwiseBT-MLE-FC, and UniformPairwiseBT-BordaPlugIn-FC. The main runs contain 3400 scaling replications and 2700 benchmark replications per algorithm across all experiment cells. Across all completed main settings, the empirical error rate is 0.0 for scaling and 0.0 for benchmarks; therefore the main comparison is sample complexity, measured by mean stopping time. The report also includes four diagnostic extensions: confidence-scaling quantiles, constant calibration, Pareto-size ablation, and correlated arena instances.

1 Protocol

The task is fixed-confidence identification of the strict Pareto set under binary pairwise comparison feedback. Each run stops adaptively when the algorithm can return a Pareto set certificate at the target failure probability. The default target is $\delta = 0.05$, except in the confidence-scaling sweep where δ itself is varied. The reported stopping time counts coordinate-specific pairwise comparison pulls. Lower stopping time is better when the empirical error rate is controlled.

All four methods use sample constant 2.0, threshold constant 4.0, and log constant 4.0. The BT-MLE baseline uses balanced random pair-coordinate allocation, ridge parameter 10^{-8} , fallback ridge parameter 10^{-4} , maximum absolute latent score 20, and MLE tolerance 10^{-9} in the benchmark run. Raw outputs are stored as CSV files because the local parquet engine was unavailable; this does not change the numerical results.

Metrics and legends. The main metric is mean stopping time $\bar{\tau}$. The normalized budget $\tau/(dH_B)$ uses the instance Borda hardness H_B and is used only to put settings with different dimensions and gaps on a comparable axis. Error plots show $\log_{10}(\max\{\hat{p}_{err}, 0.5/n\})$; because every main empirical error rate is zero, those curves sit at the finite-sample floor. In line plots, the legend entries are the four algorithm names. In bar plots, the x-axis labels are algorithm names and the y-axis is log-scaled mean stopping time. In pair-cell coverage plots, the vertical dashed line at coverage 1 means one sample per pair-coordinate cell on average.

Table 1: Algorithms compared in the fixed-confidence experiments.

Name	Sampling model	Fixed-confidence rule
VB-EGE-practical	Coordinate-wise Vector-Borda estimates	Adaptive accept/reject using empirical gap certificates
UniformFocalBorda-FC	Uniform coordinate-wise Borda sampling	Same confidence threshold family without adaptive elimination
UniformPairwiseBT-MLE-FC	Uniform pair-coordinate Bradley-Terry sampling with MLE	Stop when plug-in Pareto set is certified by BT confidence radius
UniformPairwiseBT-BordaPlugIn-FC	Uniform pair-coordinate sampling with Borda plug-in	Stop when Borda plug-in certificate is satisfied

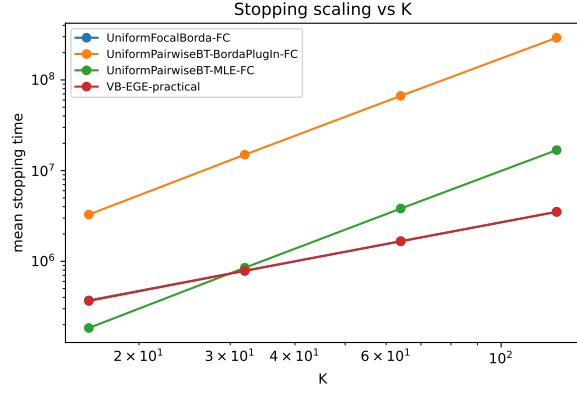
Table 2: Fixed-confidence scaling settings. All tasks use the symmetric hard generator and four algorithms; each cell is averaged over 200 random permutations/seeds.

Sweep	Generator	Grid	Purpose
K scaling	symmetric_hard	K in {16,32,64,128}; d=4; Delta=1; delta=0.05	One Pareto arm at the origin and K-1 dominated arms at -Delta in every coordinate.
d scaling	symmetric_hard	K=64; d in {2,4,10}; Delta=1; delta=0.05	Symmetric hard instance with the number of dimensions varied.
Gap scaling	symmetric_hard	K=64; d=10; Delta in {0.5,0.75,1,1.25,1.5}; delta=0.05	Symmetric hard instance with the latent separation Delta varied.
Confidence scaling	symmetric_hard	K=64; d=10; Delta=1; delta in {0.20,0.10,0.05,0.02,0.01}	Symmetric hard instance with the target failure probability varied.

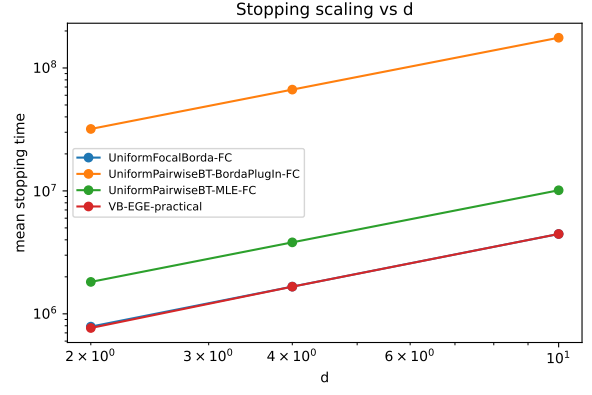
2 Fixed-Confidence Scaling Tasks

The scaling tasks isolate how stopping time changes with the number of arms, the dimension, the latent gap, and the confidence level on the symmetric hard instance. In this generator, there is a single true Pareto arm and all other arms are dominated by the same coordinate gap. This makes the setting useful for checking whether the fixed-confidence rules behave monotonically and whether pairwise baselines pay a large cost for broad pair-coordinate coverage.

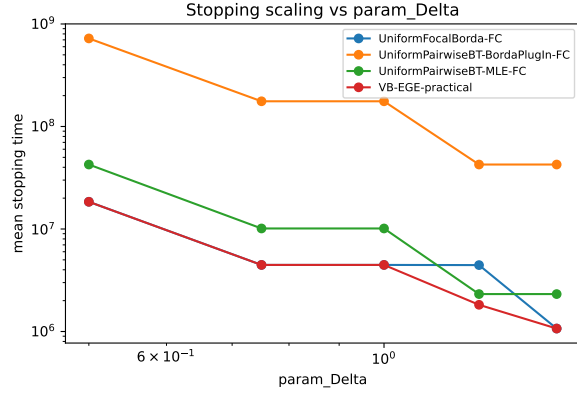
Result interpretation. VB-EGE-practical and UniformFocalBorda-FC are nearly tied on the symmetric hard instance, as expected because the instance has a single symmetric Pareto arm and little need for adaptive frontier structure. The pairwise BT-MLE and Borda plug-in baselines are substantially slower as K and d grow, reflecting the cost of spreading samples across many pair-coordinate cells. Larger gaps reduce stopping time, while smaller target delta increases stopping time through the confidence threshold. The small plateaus in the gap sweep are consistent with dyadic phase schedules: once two gap values trigger the same final phase, their stopping times can match even if the underlying gap differs.



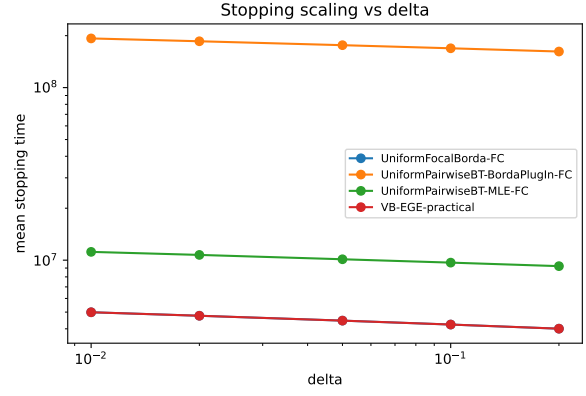
(a) K scaling



(b) d scaling



(c) Gap scaling



(d) Confidence scaling

Figure 1: Scaling curves for mean stopping time. Each line is one algorithm. The axes are log-scaled; lower curves indicate fewer comparison pulls.

Table 3: Scaling results. For each swept parameter value, the table reports VB-EGE mean stopping time and baseline ratios relative to VB-EGE. All empirical error rates are zero.

Sweep	Value	VB-EGE $\bar{\tau}$	Focal/VB	MLE/VB	Plug-in/VB
K scaling	16	3.65×10^5	1.02x	0.50x	8.97x
K scaling	32	7.84×10^5	1.00x	1.08x	19.1x
K scaling	64	1.66×10^6	1.00x	2.29x	40.1x
K scaling	128	3.51×10^6	1.00x	4.80x	83.2x
d scaling	2	7.67×10^5	1.03x	2.37x	41.6x
d scaling	4	1.66×10^6	1.00x	2.29x	40.1x
d scaling	10	4.46×10^6	1.00x	2.27x	39.5x
Gap scaling	0.50	1.84×10^7	1.00x	2.31x	39.2x
Gap scaling	0.75	4.46×10^6	1.00x	2.27x	39.5x
Gap scaling	1	4.46×10^6	1.00x	2.27x	39.5x
Gap scaling	1.25	1.82×10^6	2.44x	1.27x	23.4x
Gap scaling	1.50	1.07×10^6	1.00x	2.17x	39.8x
Confidence scaling	0.01	4.99×10^6	1.00x	2.24x	38.6x
Confidence scaling	0.02	4.76×10^6	1.00x	2.25x	39.0x
Confidence scaling	0.05	4.46×10^6	1.00x	2.27x	39.5x
Confidence scaling	0.10	4.23×10^6	1.00x	2.29x	39.9x
Confidence scaling	0.20	4.01×10^6	1.00x	2.30x	40.4x

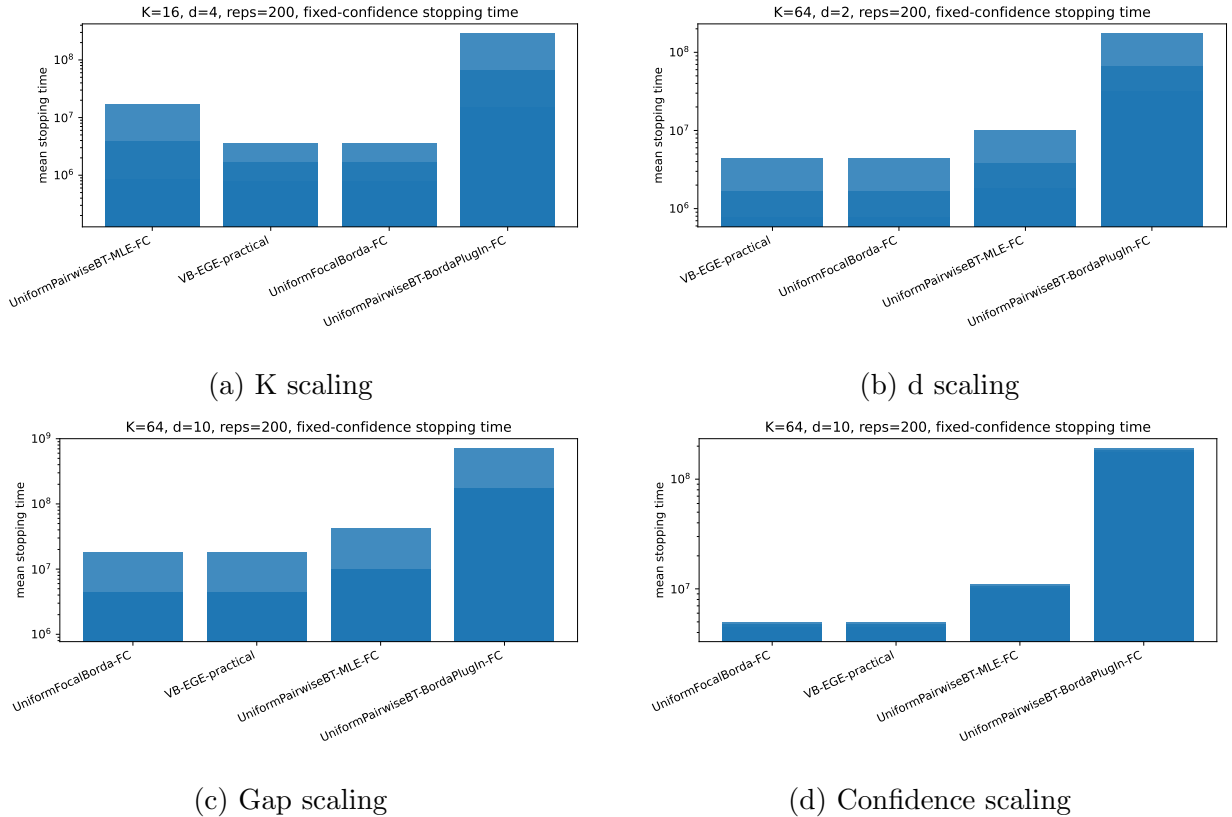
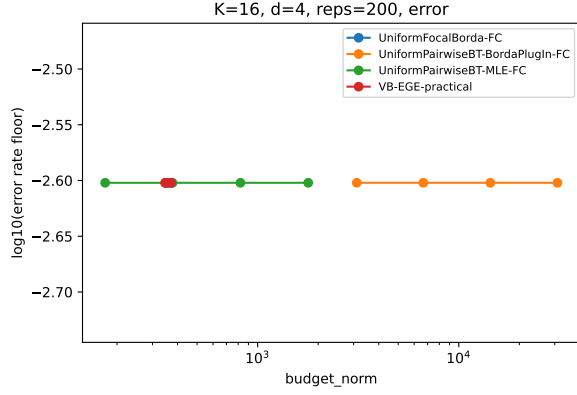
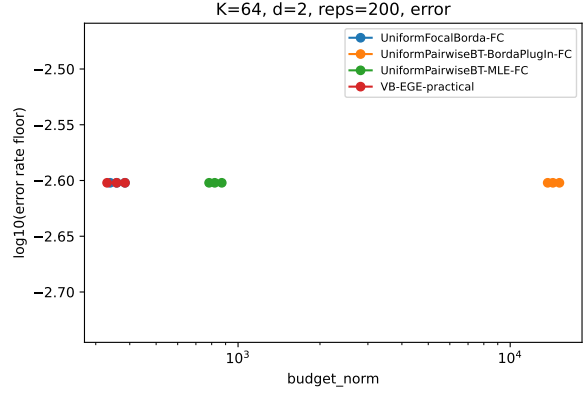


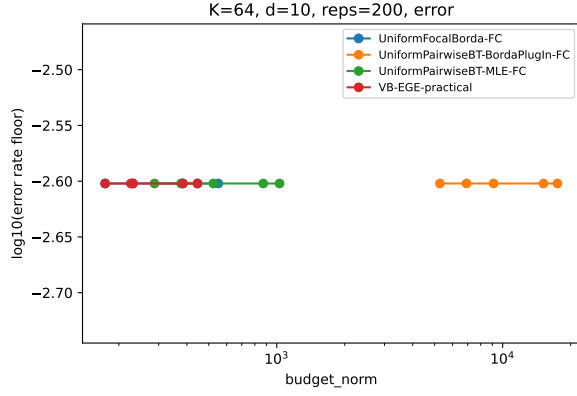
Figure 2: Per-sweep fixed-confidence stopping bars. Bars are sorted by mean stopping time within each plotted setting and use a log y-axis.



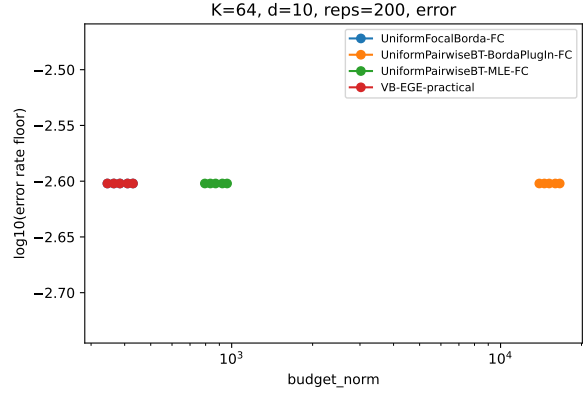
(a) K scaling



(b) d scaling



(c) Gap scaling



(d) Confidence scaling

Figure 3: Scaling error-rate floors under normalized budget. All empirical errors are zero, so the curves show the floor $0.5/n$ rather than observed failures.

Table 4: Fixed-confidence benchmark settings. All algorithms use target failure probability $\delta = 0.05$; $|P|$ denotes the true Pareto set size.

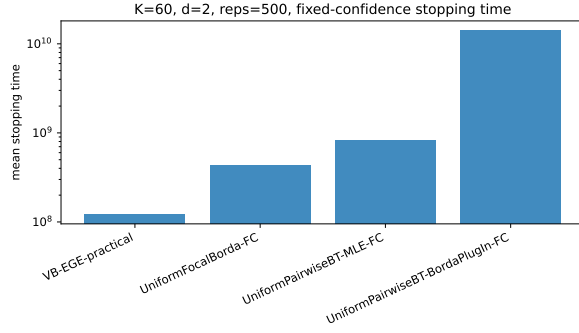
Setting	Generator	K	d	$ P $	Reps	Parameters
Convex-2D	convex_frontier_2d	60	2	15	500	s=15
Arena-4 small	arena_tradeoff_frontier	32	4	8	500	s=8, margins [0.08,0.25], alpha=0.7
Arena-4 medium	arena_tradeoff_frontier	64	4	12	300	s=12, margins [0.08,0.25], alpha=0.7
Arena-10 medium	arena_tradeoff_frontier	64	10	16	300	s=16, margins [0.06,0.20], alpha=0.7
Witness-4	unique_witness_d	40	4	8	500	s=8, q=4, margins [0.04,0.16]
Witness-10	unique_witness_d	80	10	16	300	s=16, q=4, margins [0.035,0.14]
Two-group-10	highdim_two-group	64	10	15.79	300	K_low=48, K_high=16

Table 5: Benchmark results. The table reports the mean stopping time of VB-EGE and the multiplicative stopping-time ratio of each baseline relative to VB-EGE. All empirical error rates are zero.

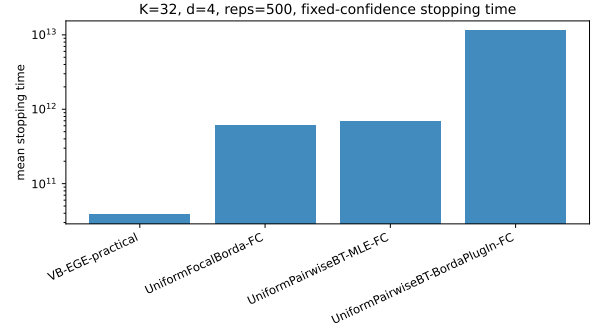
Setting	VB-EGE $\bar{\tau}$	Focal/VB	MLE/VB	Plug-in/VB	Wilson upper
Convex-2D	1.21×10^8	3.55x	6.90x	118x	0.0076
Arena-4 small	3.83×10^{10}	15.7x	18.3x	302x	0.0076
Arena-4 medium	3.65×10^{10}	31.6x	69.1x	1227x	0.0126
Arena-10 medium	9.79×10^8	13.0x	27.6x	500x	0.0126
Witness-4	2.28×10^8	1.25x	1.79x	29.7x	0.0076
Witness-10	1.61×10^9	2.15x	2.95x	48.7x	0.0126
Two-group-10	7.16×10^{11}	115x	270x	4359x	0.0126

3 Fixed-Confidence Benchmark Tasks

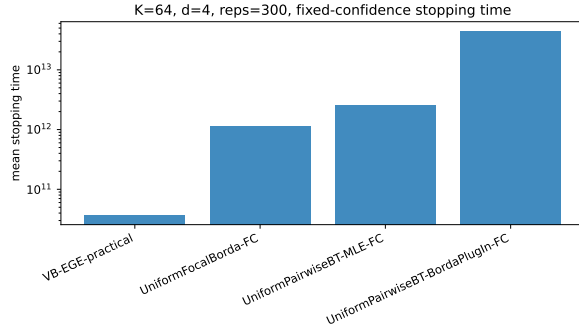
The benchmark suite stresses different Pareto geometries beyond the symmetric hard case. Convex-2D tests a clean low-dimensional trade-off frontier. Arena settings use mutually non-dominated anchors and dominated alternatives offset from a random anchor. Witness settings create dominated arms that are certified by a unique Pareto witness, which tests whether algorithms can avoid unnecessary global pairwise coverage. Two-group-10 is a high-dimensional separation task with many low-quality arms and a smaller high-quality group.



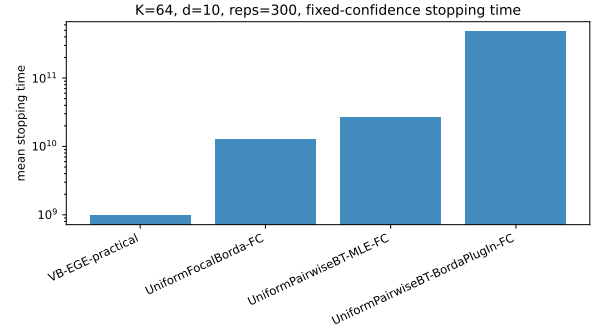
(a) Convex-2D



(b) Arena-4 small

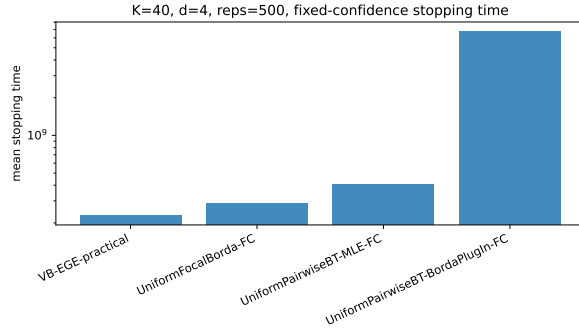


(c) Arena-4 medium

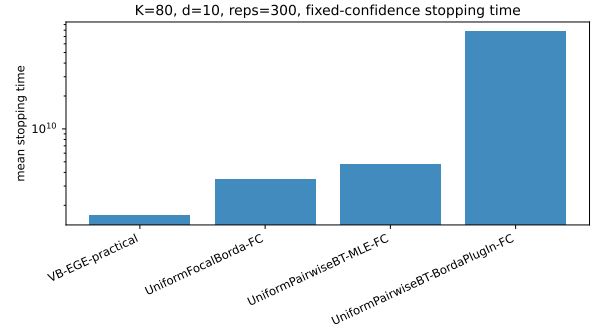


(d) Arena-10 medium

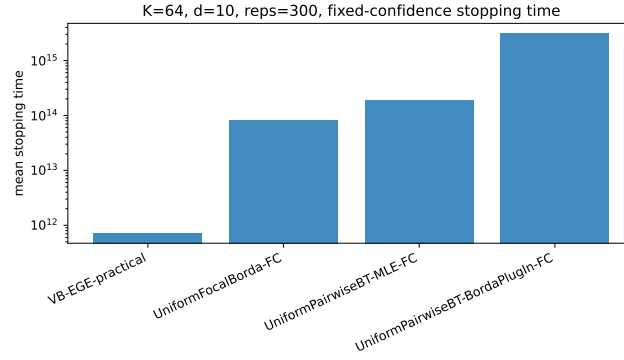
Figure 4: Benchmark stopping times, first group. Each bar is one algorithm; lower bars mean fewer comparison pulls before certification.



(a) Witness-4

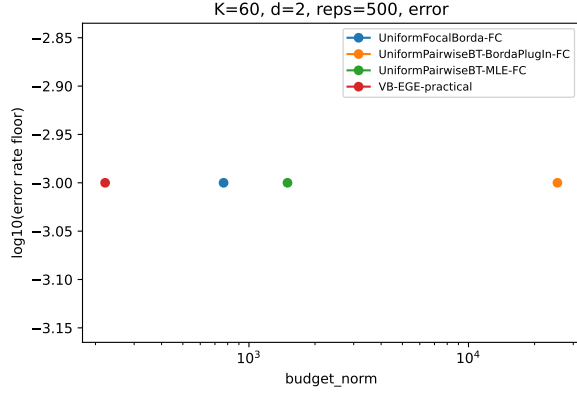


(b) Witness-10

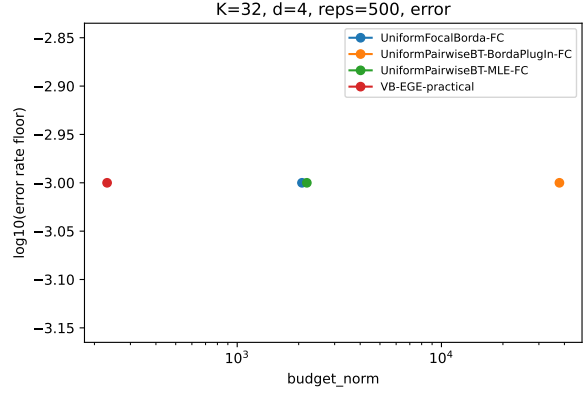


(c) Two-group-10

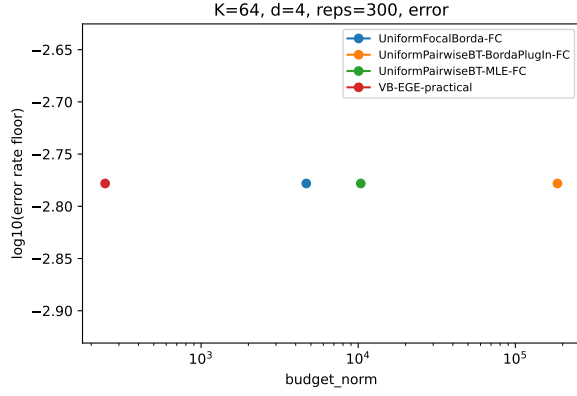
Figure 5: Benchmark stopping times, second group. These tasks emphasize local witnesses and high-dimensional group separation.



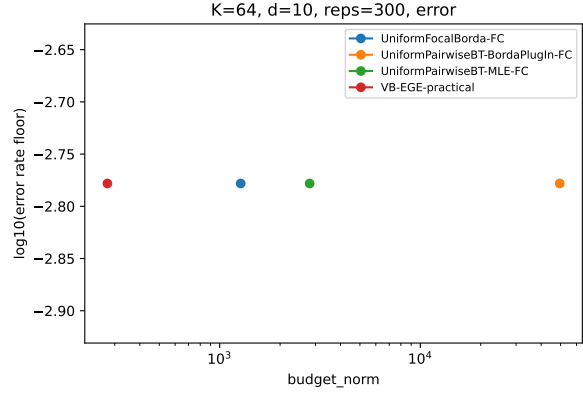
(a) Convex-2D



(b) Arena-4 small

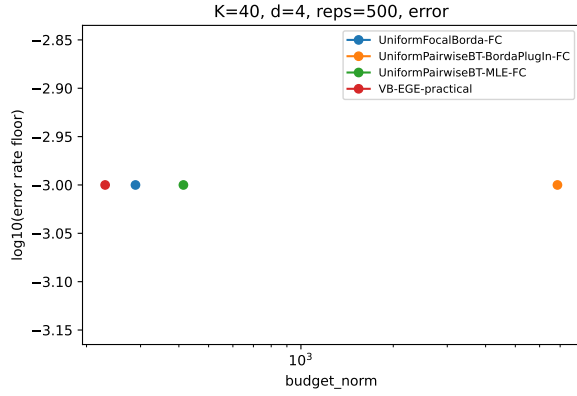


(c) Arena-4 medium

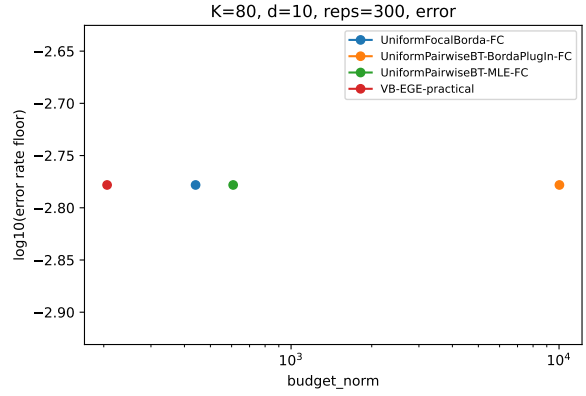


(d) Arena-10 medium

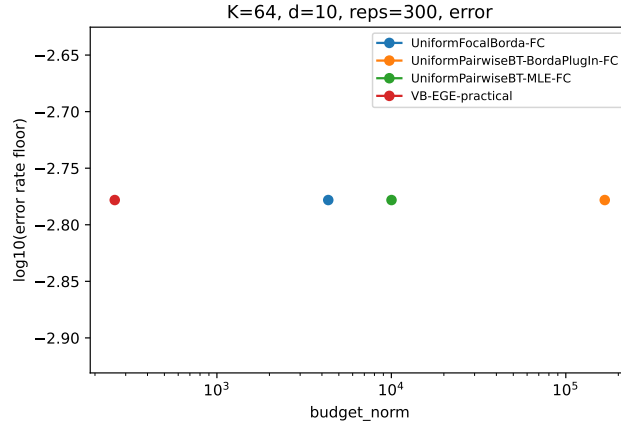
Figure 6: Benchmark normalized error floors, first group. All empirical errors are zero; the y-values are finite-sample floors.



(a) Witness-4



(b) Witness-10



(c) Two-group-10

Figure 7: Benchmark normalized error floors, second group. Curves at the same floor indicate exact empirical recovery across all replications.

Result interpretation. VB-EGE-practical has the smallest mean stopping time in every benchmark setting while maintaining zero empirical error. The advantage is moderate on unique-witness instances, where the focal Borda baseline already receives local certificate information, and much larger on arena and two-group instances. The BT-MLE baseline has near-perfect pairwise sign accuracy in the completed runs, but it pays a large exploration cost before enough pair-coordinate cells are covered. The Borda plug-in pairwise baseline is consistently the slowest, which suggests that uniform pairwise sampling is poorly matched to fixed-confidence Pareto certification when the certificate can be expressed in the Vector-Borda embedding.

Table 6: Confidence-scaling quantile settings. These runs use the same fixed-confidence algorithms as the main benchmark suite and vary only the requested failure probability.

Setting	Generator	Grid	Purpose	Reps
Symmetric K64 d10	symmetric_hard	K=64; d=10; Delta=1; delta in {0.20,0.10,0.05,0.02,0.01,0.005,0.002,0.001}	Stress the tail of fixed-confidence stopping times as $\log(1/\delta)$ grows.	500
Arena-10 medium	arena_tradeoff_frontier	K=64; d=10; s=16; margins [0.06,0.20]; delta in {0.20,0.10,0.05,0.02,0.01,0.005}	Check confidence scaling on a high-dimensional multi-Pareto arena.	300

4 Additional Fixed-Confidence Synthetic Tasks

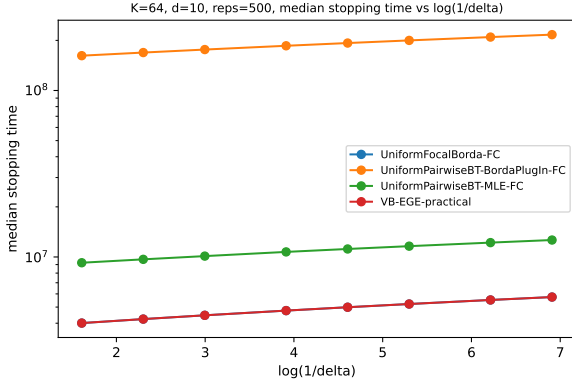
4.1 Confidence-Scaling Quantiles

This extension repeats fixed-confidence runs over a wider grid of target failure probabilities. The line legends are the four algorithms; the shaded or separated quantile traces report the median and upper stopping-time quantiles. The x-axis is $\log(1/\delta)$, so approximately linear growth indicates the expected logarithmic confidence cost. The normalized panels divide by the instance Borda hardness, and the $\tau/\log(1/\delta)$ panels make deviations from pure logarithmic scaling visible.

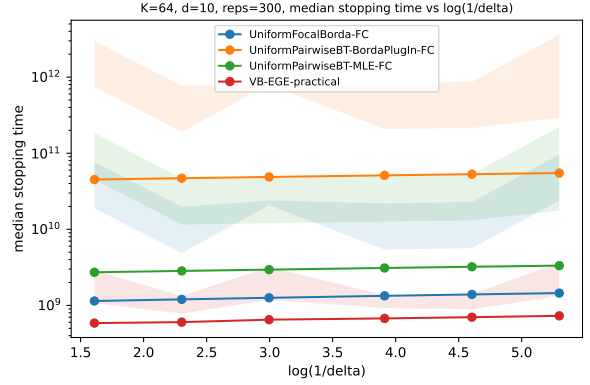
Result interpretation. On the symmetric setting, VB-EGE and the focal Borda baseline remain close because the certificate is essentially one-dimensional after symmetry. The arena setting separates the methods more clearly: VB-EGE keeps a smaller high quantile stopping time than the pairwise baselines because it does not need broad pair-coordinate coverage at every confidence level. The final-phase diagnostic should move in steps rather than smoothly, which is expected from phase doubling.

Table 7: Confidence-scaling quantile results. VB columns report mean and 95th-percentile stopping times; ratios use mean stopping time relative to VB-EGE.

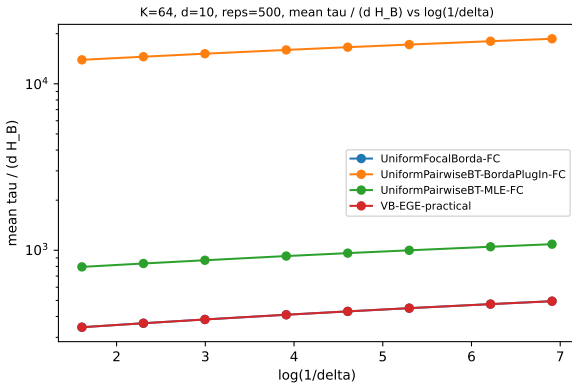
Setting	δ	VB mean τ	VB q95 τ	Focal/VB	MLE/VB	Plug-in/VB
Symmetric K64 d10	0.200	4.01×10^6	4.01×10^6	1.00x	2.30x	40.4x
Symmetric K64 d10	0.100	4.23×10^6	4.23×10^6	1.00x	2.29x	39.9x
Symmetric K64 d10	0.050	4.46×10^6	4.46×10^6	1.00x	2.27x	39.5x
Symmetric K64 d10	0.020	4.76×10^6	4.76×10^6	1.00x	2.25x	39.0x
Symmetric K64 d10	0.010	4.99×10^6	4.99×10^6	1.00x	2.24x	38.6x
Symmetric K64 d10	0.005	5.22×10^6	5.22×10^6	1.00x	2.23x	38.3x
Symmetric K64 d10	0.002	5.52×10^6	5.52×10^6	1.00x	2.21x	38.0x
Symmetric K64 d10	0.001	5.74×10^6	5.74×10^6	1.00x	2.20x	37.7x
Arena-10 medium	0.200	3.33×10^9	2.88×10^9	27.1x	61.8x	1044x
Arena-10 medium	0.100	2.11×10^9	1.33×10^9	23.4x	40.8x	907x
Arena-10 medium	0.050	6.14×10^{11}	3.19×10^9	32.0x	75.1x	1214x
Arena-10 medium	0.020	6.48×10^{11}	1.37×10^9	32.0x	74.5x	1204x
Arena-10 medium	0.010	1.39×10^9	1.41×10^9	17.2x	38.8x	657x
Arena-10 medium	0.005	4.55×10^{10}	3.64×10^9	31.5x	72.6x	1176x



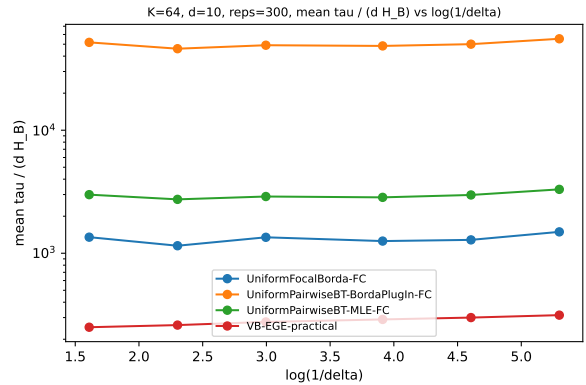
(a) Symmetric K64 d10



(b) Arena-10 medium

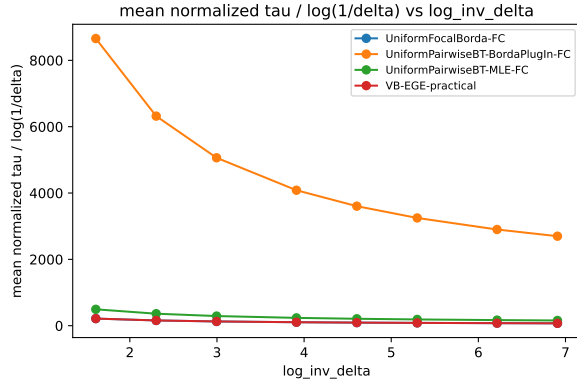


(c) Normalized symmetric

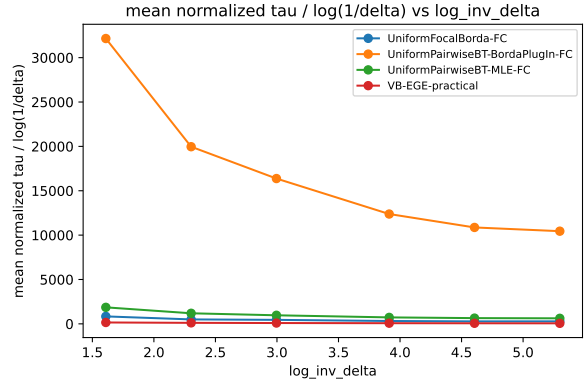


(d) Normalized arena10

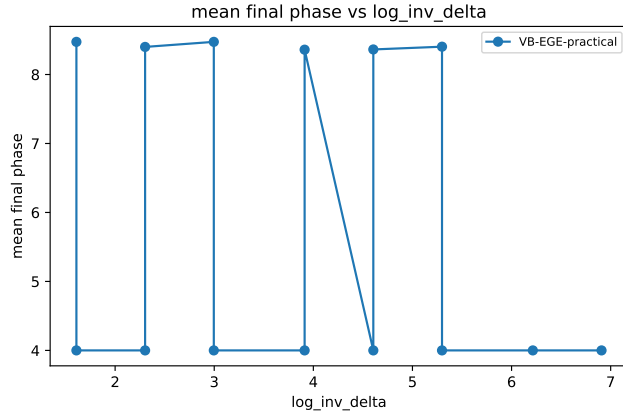
Figure 8: Confidence-scaling stopping quantiles. Each color is one algorithm; lower median and q95 traces mean fewer pulls at the same confidence target.



(a) Symmetric tau/log



(b) Arena10 tau/log



(c) VB final phase

Figure 9: Confidence-scaling diagnostics. Stable tau/log curves support logarithmic dependence on $1/\delta$; the final-phase plot shows the dyadic phase schedule used by VB-EGE.

Table 8: Constant-calibration settings. VB-EGE is swept over sample constants $\{0.5, 1, 2, 4\}$ and threshold constants $\{2, 3, 4, 6\}$ at $\delta = 0.05$; the focal and BT-MLE references use the default constants (2, 4).

Setting	Generator	Parameters	Grid	Reps
Symmetric K64 d10	symmetric_hard	K=64; d=10; Delta=1	$c_s \in \{0.5, 1, 2, 4\}; c_\theta \in \{2, 3, 4, 6\}$	500
Arena-10 medium	arena_tradeoff_frontier	K=64; d=10; s=16; margins [0.06,0.20]	$c_s \in \{0.5, 1, 2, 4\}; c_\theta \in \{2, 3, 4, 6\}$	300
Witness-10	unique_witness_d	K=80; d=10; s=16; q=4; margins [0.035,0.14]	$c_s \in \{0.5, 1, 2, 4\}; c_\theta \in \{2, 3, 4, 6\}$	300

4.2 Constants Calibration

The calibration sweep varies the VB-EGE sample constant c_s and threshold constant c_θ . In the frontier figures, each point is one constant pair; the x-axis is mean stopping time and the y-axis is either empirical error or Wilson upper confidence bound. Heatmaps use sample constant on one axis and threshold constant on the other; darker or lighter cells encode the plotted metric named in the title. The default paper setting is $(c_s, c_\theta) = (2, 4)$.

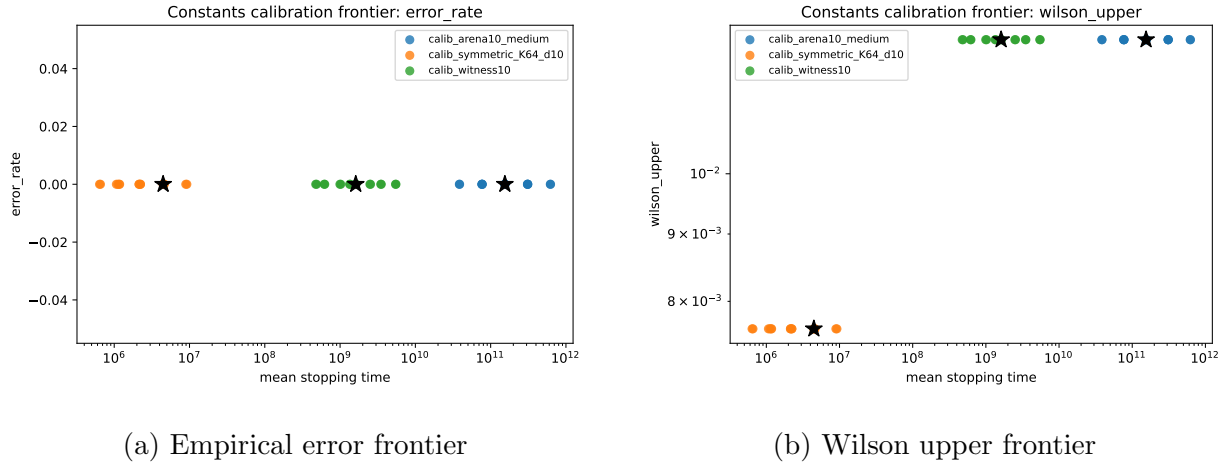
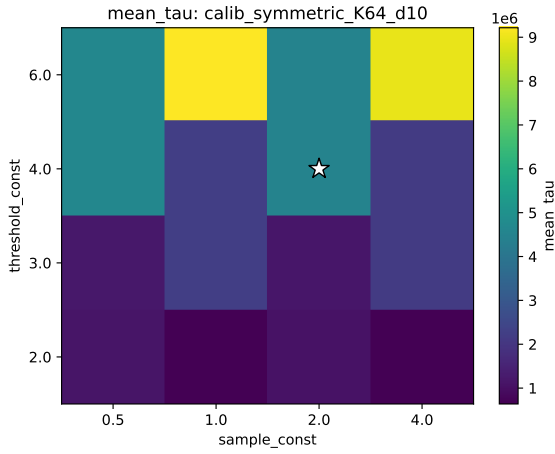


Figure 10: Calibration frontiers. Points near the lower-left are better only if their error remains below the target confidence level.

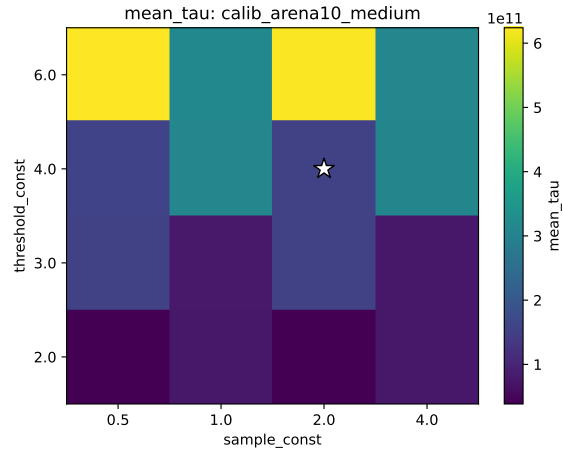
Result interpretation. The calibration task is not intended to declare every constant pair safe; it shows the efficiency-reliability frontier. Smaller constants can reduce stopping time, but only constants with acceptable empirical error and complete stopping are viable. The default constants are therefore a conservative reference point for the main benchmark comparisons.

Table 9: Constant-calibration results. The default row uses $(c_s, c_\theta) = (2, 4)$. The best row is the smallest mean stopping time among grid points with empirical error at most 0.05 and complete stopping.

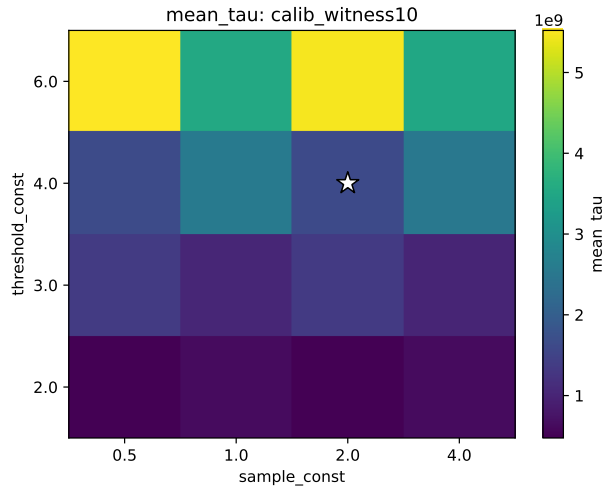
Setting	Default VB $\bar{\tau}$	Default err	Best constants	Best VB $\bar{\tau}$	Focal/default	MLE/default
Symmetric K64 d10	4.46×10^6	0.000	(4, 2)	6.38×10^5	1.00x	2.27x
Arena-10 medium	1.55×10^{11}	0.000	(2, 2)	3.84×10^{10}	31.9x	74.9x
Witness-10	1.62×10^9	0.000	(2, 2)	4.75×10^8	2.16x	2.94x



(a) Symmetric K64 d10



(b) Arena-10 medium



(c) Witness-10

Figure 11: Calibration mean stopping-time heatmaps. Each cell is one VB-EGE constant pair; the color scale is mean stopping time.

Table 10: Pareto-size ablation settings. The arena generator is held fixed while the target frontier size $s = |P|$ changes.

Setting	Generator	Grid	Purpose	Reps
Arena-4	arena_tradeoff_frontier	K=64; d=4; s in {4,8,16,32}; margins [0.08,0.25]	Measure how certificate cost changes as the frontier grows.	300
Arena-10	arena_tradeoff_frontier	K=64; d=10; s in {4,8,16,32}; margins [0.06,0.20]	Measure how certificate cost changes as the frontier grows.	300

4.3 Pareto-Size Ablation

This ablation holds the arena geometry fixed while increasing the true Pareto frontier size $|P|$. In the plots, x-axis values are the target frontier sizes and each legend entry is an algorithm. The baseline-ratio plot reports mean stopping time divided by VB-EGE at the same setting; values above one favor VB-EGE.

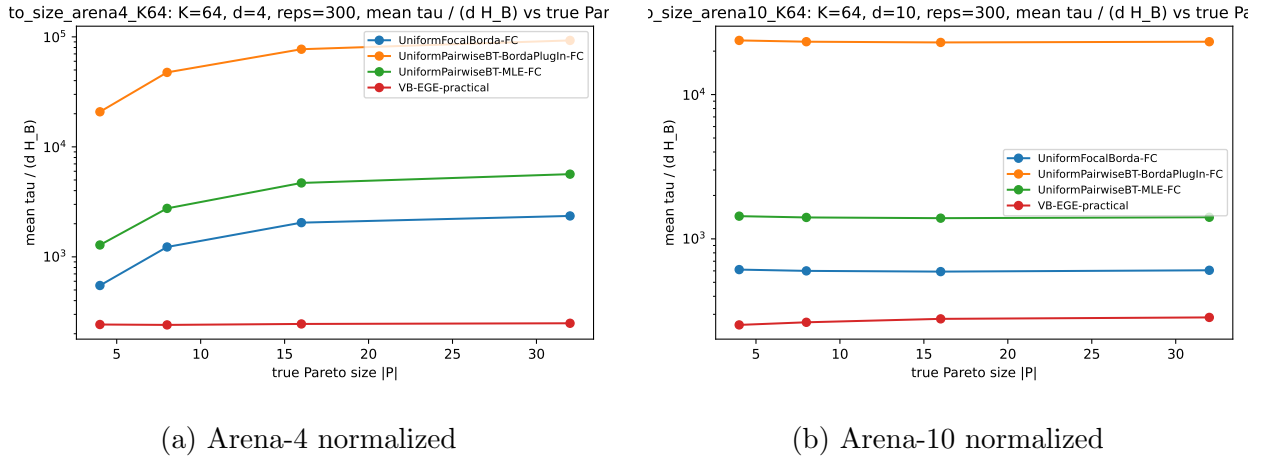


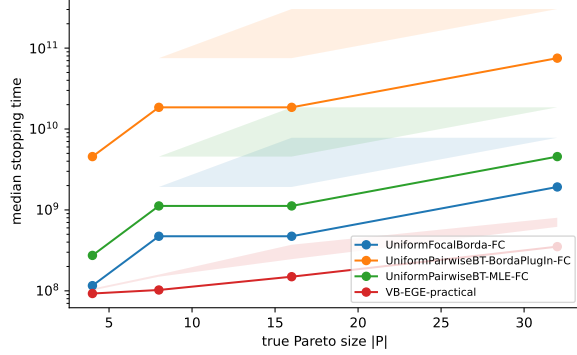
Figure 13: Pareto-size normalized stopping times. Normalization by dH_B separates hardness changes from pure frontier-size effects.

Result interpretation. The important question is whether the relative advantage of VB-EGE persists as the Pareto frontier grows. When the ratio curves stay above one, the adaptive Vector-Borda certificate remains more sample-efficient than uniform baselines even though more arms need to be accepted.

Table 11: Pareto-size ablation results. Ratios are mean stopping-time ratios relative to VB-EGE at the same arena dimension and target Pareto size.

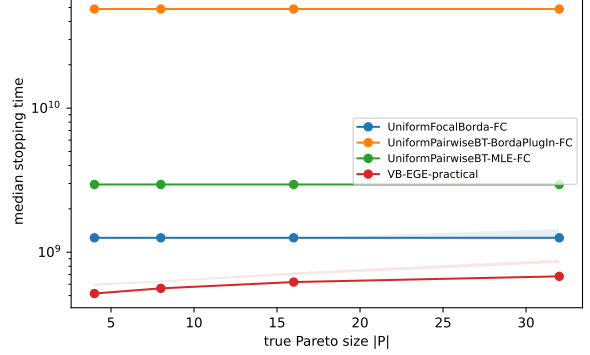
Setting	$ P $	VB $\bar{\tau}$	VB $\tau/(dH_B)$	Focal/VB	MLE/VB	Plug-in/VB
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_size_arena4_K64: K=64, d=4, reps=300, median stopping time vs true P

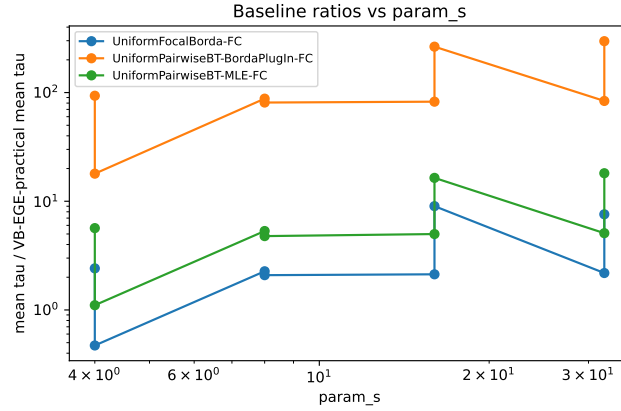


(a) Arena-4 tau

_size_arena10_K64: K=64, d=10, reps=300, median stopping time vs true P



(b) Arena-10 tau



(c) Baseline/VB ratio

Figure 12: Pareto-size stopping-time ablation. Growing $|P|$ increases the number of arms that must be accepted and changes the number of dominated arms left to reject.

Table 12: Correlated-arena settings. The correlated generator constructs frontier anchors from shared latent factors and records the achieved average off-diagonal objective correlation.

Setting	Generator	Grid	Purpose	Reps
Correlated Arena-4	correlated_arena_like	K=32; d=4; s=8; rho in {0,0.6,0.9}	Test whether correlated objectives change sample complexity.	300
Correlated Arena-10	correlated_arena_like	K=64; d=10; s=16; rho in {0,0.3,0.6,0.85,0.95}	Test whether correlated objectives change sample complexity.	300

4.4 Correlated Arena

The correlated arena generator adds a target objective-correlation parameter ρ . The achieved-correlation column reports the actual mean off-diagonal correlation after resampling instances that have the requested Pareto size. Line legends again identify the algorithms. The pair-cell coverage diagnostic is meaningful only for pairwise baselines, while the Borda-gap plot reports the average minimum Borda certificate gap.

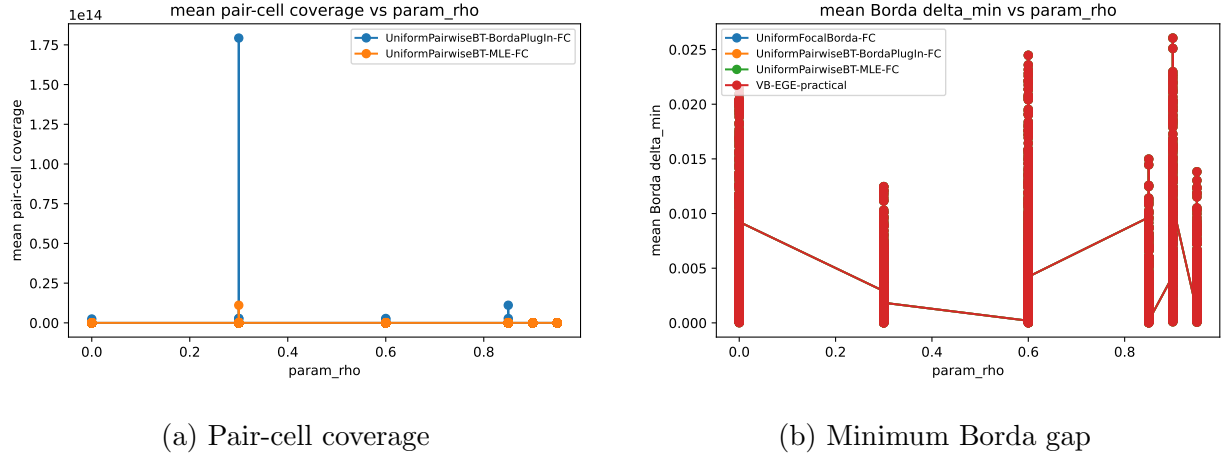


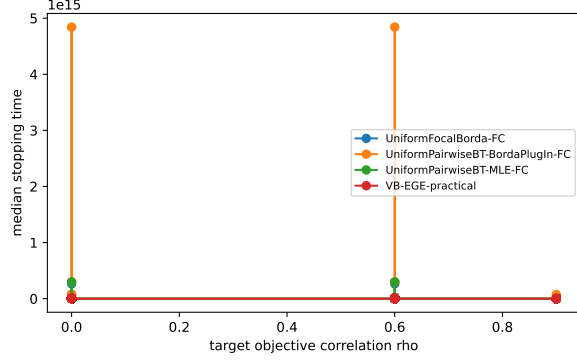
Figure 15: Correlated-arena diagnostics. These diagnostics help distinguish a sampling-coverage effect from a genuine hardness change in the Borda certificate.

Result interpretation. Correlation changes both the geometry of the objectives and the effective Borda gaps. If stopping time increases while the minimum Borda gap shrinks, the harder certificate explains the cost. If pairwise baselines grow faster than VB-EGE at similar Borda gaps, the extra cost comes from their broad direct pair-coordinate sampling requirement.

Table 13: Correlated-arena results. Achieved corr. is the mean off-diagonal objective correlation produced by the generator.

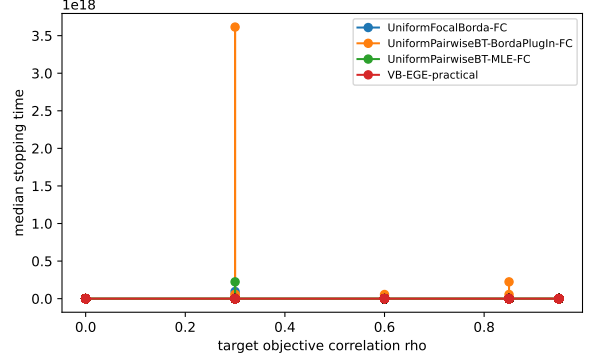
Setting	Target ρ	Achieved corr.	VB $\bar{\tau}$	VB $\tau/(dH_B)$	Focal/VB	MLE/VB	Plug-in/VB
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na4_K32: K=32, d=4, reps=1, median stopping time vs target objective cc

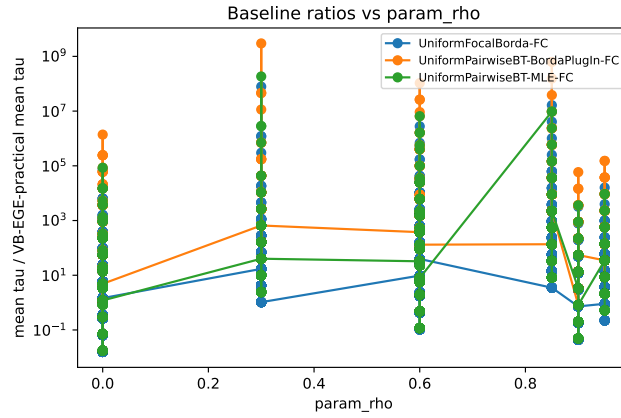


(a) Arena-4 tau

na10_K64: K=64, d=10, reps=2, median stopping time vs target objective



(b) Arena-10 tau



(c) Baseline/VB ratio

Figure 14: Correlated-arena stopping times. The x-axis is target ρ ; lower curves and higher baseline/VB ratios favor VB-EGE.

5 Pair-Cell Coverage Diagnostics

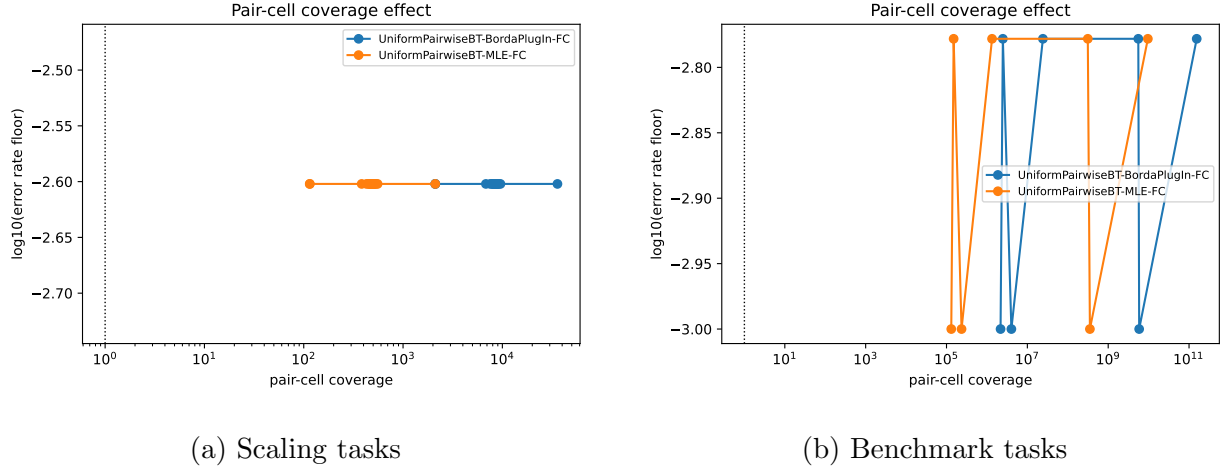


Figure 16: Pair-cell coverage diagnostics. Only pairwise baselines appear in these diagnostics. Coverage is total sampled pair-coordinate cells divided by $dK(K-1)/2$.

The coverage diagnostic explains a major source of the stopping-time gap. Pairwise baselines must collect enough direct pair-coordinate evidence to certify many signs or plug-in Borda gaps. VB-EGE instead estimates coordinate-wise Vector-Borda scores and can stop after certifying accepted and rejected arms in that lower-dimensional certificate space.

6 Takeaways

1. All completed main fixed-confidence synthetic experiments achieved zero empirical Pareto-set error.
2. VB-EGE-practical is the most sample-efficient method on every benchmark setting and is competitive with the focal Borda baseline on the symmetric hard scaling task.
3. Pairwise BT-MLE can estimate latent pairwise signs accurately, but fixed-confidence Pareto certification makes its broad pair-coordinate coverage expensive.
4. The largest gains occur in high-dimensional arena and two-group settings, where adaptive Vector-Borda certificates avoid many unnecessary pairwise comparisons.

Reproducibility Notes

The source summaries used in this report are `results/summary/fixed_confidence_scaling_summary.csv` and `results/summary/fixed_confidence_benchmarks_summary.csv`, plus `results/summary/confidence_scaling_quantile_summary.csv`, `results/summary/constants_calibration_summary.csv`, `results/summary/pareto_size_ablation_summary.csv`, and `results/summary/correlated_arena_summary.csv`. The raw completed main outputs are `results/raw/fixed_`

`confidence_scaling.csv`, `results/raw/fixed_confidence_benchmarks.csv`, and the corresponding CSV fallbacks under `results/raw/` for the four diagnostic extensions. Smoke runs are excluded from the main conclusions because they were used only to validate the implementation path before the full runs.